Cryptography and Security Implementations

RC4 Assignment

Assignment

Implement the **RC4 stream cipher** in the C programming language.

Tasks

1. Implement the **Key Scheduling Algorithm (KSA)**.
2. Implement the **Pseudo-Random Generation Algorithm (PRGA)**.
3. Measure and report the **total clock cycles** required by the KSA.
4. Measure and report the **clock cycles per output byte** for the PRGA.
5. Submit:
    - Source code.
    - Experimental measurements.
    - A brief analysis explaining the observed measurements.

Deadline: Today (11:59 PM)

Reference Algorithms

The following algorithms are reproduced from:

> Dan Boneh and Victor Shoup, *A Graduate Course in Applied Cryptography*, RC4 Stream Cipher.

Algorithm 1 RC4 Key Scheduling Algorithm (KSA)

Require: Seed (key) consisting of s bytes

$S[i] \leftarrow i$ for $i = 0, \ldots, 255$
$j \leftarrow 0$
for $i = 0$ to 255 **do**
    $k \leftarrow \text{key}[\, i \bmod s \,]$
    $j \leftarrow (j + S[i] + k) \bmod 256$
    $\text{Swap}(S[i], S[j])$
end for

Algorithm 2 RC4 Pseudo-Random Generation Algorithm (PRGA)

1: $i \leftarrow 0$
2: $j \leftarrow 0$
3: **while** more output bytes are required **do**
4: $\quad i \leftarrow (i + 1) \bmod 256$
5: $\quad j \leftarrow (j + S[i]) \bmod 256$
6: $\quad \text{Swap}(S[i], S[j])$
7: $\quad \text{Output } S[(S[i] + S[j]) \bmod 256]$
8: **end while**